

TW01: Foundations of Data Science

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Start from zero by working with values, variables, calculations, text, comparisons, decisions, and a small business question.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Create your first Python data card

Scenario

You are joining a data team and need a short program that stores and describes four different kinds of values.

INSTRUCTIONS

1. Create variables for a student name, training week, room temperature, and whether the class is active.
2. Use a meaningful variable name for every value.
3. Print each value and its data type using `type()`.
4. Add at least two comments that explain what the code does.
5. Write one sentence explaining the difference between text, whole numbers, decimal numbers, and Boolean values.

REQUIRED SUBMISSION

- A working code cell containing four variables.
- Four labelled values and four displayed data types.
- Two useful comments and one plain-language explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Build a simple sales receipt

Scenario

A small shop wants a transparent calculation for a customer's purchase.

INSTRUCTIONS

1. Store the product name, unit price, quantity, and tax rate in variables.
2. Calculate subtotal, tax amount, and final total.
3. Print a readable receipt with labels and currency values rounded to two decimal places.
4. Change the quantity and rerun the cell to prove the program updates automatically.
5. Add one comparison that checks whether the final total is more than 50 pounds.

REQUIRED SUBMISSION

- A receipt containing product, quantity, subtotal, tax, and total.
- A Boolean result for the 50-pound comparison.
- A short note describing why variables are better than typing totals manually.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Standardise and inspect text

Scenario

Product names have arrived with extra spaces and inconsistent capital letters.

INSTRUCTIONS

1. Remove leading and trailing spaces from the supplied text.
2. Create lowercase, uppercase, and title-case versions.
3. Count the number of characters after spaces are removed.
4. Check whether the cleaned text contains the word data.
5. Create a sentence using an f-string that reports the cleaned product and its length.

REQUIRED SUBMISSION

- Four clearly labelled transformed text values.
- A correct character count and membership check.
- One readable f-string summary.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Make a delivery decision

Scenario

An online shop offers free delivery when the basket total reaches a stated threshold.

INSTRUCTIONS

1. Set a basket total and free-delivery threshold.
2. Use if and else to create a suitable customer message.
3. If delivery is not free, calculate how much more the customer must spend.
4. Test the program with one value below and one value at or above the threshold.
5. Explain why the boundary value must be tested.

REQUIRED SUBMISSION

- A working if/else decision.
- Output from two different test cases.
- A one-sentence boundary-test explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Answer a first business question

Scenario

A cafe manager has seven daily sales totals and wants a short evidence-based summary.

INSTRUCTIONS

1. Calculate the number of days, total sales, average sales, lowest sales, and highest sales using built-in functions.
2. Compare average sales with the target of 500 pounds.
3. Use if/else to create a message stating whether the target was achieved.
4. Identify the data as structured, semi-structured, or unstructured and justify the choice.
5. Write two sentences: one finding and one limitation.

REQUIRED SUBMISSION

- Five labelled summary measures.
- A correct target decision.
- A data-type classification, finding, and limitation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____